Step 2b - H9 - Electrified mineral drying

Tech baseline

Technologies used in the market

Direct-fired rotary dryer (incumbent)

How it works: Gas/oil burner heats air blown through a rotating drum; co- or counter-current contact evaporates moisture from bulk solids.

How well it performs: ~3,500-9,000 kJ/kg water (~1.0-2.5 kWh/kg); BEE field data 788-1,416 kcal/kg; units 20-100+ t/h; ~30-60% thermal efficiency vs 0.63 kWh/kg latent floor.

Limits: Fossil heat source with exhaust losses; EU ETS/MCPD burden; 'electrifying' it means bolting on resistive/H2 burners with unchanged air losses.

Fluid-bed dryer (incumbent)

How it works: Hot gas fluidizes fine particles through a perforated plate/distributor for high heat/mass transfer.

How well it performs: ~4,000-6,000 kJ/kg water (~1.1-1.7 kWh/kg); single units to ~64-128 t/h on ore concentrates (e.g. Asarco copper).

Limits: Needs fluidizable particle size; high fan power; still gas-fired air heaters; fines entrainment/dust load.

Flash/pneumatic dryer (incumbent)

How it works: Wet fines dispersed in a hot gas stream, drying in seconds; often integrated in the grinding mill circuit.

How well it performs: Very high evaporation flux (~750 kg/m2h); e.g. Mondo Minerals talc flash drying at ~450C air, ~20 GWh/yr site demand.

Limits: Surface moisture on fine powders only; high exhaust-volume losses; fossil-fired today.

Spray dryer (incumbent, slurries)

How it works: Kaolin/mineral slurry atomized into hot air to make dry powder directly.

How well it performs: 4,500-11,500 kJ/kg water (~1.3-3.2 kWh/kg) — the most energy-hungry dryer type; standard for slurry feeds.

Limits: Only viable for pumpable slurries; huge air heating duty makes electrification hard except via SHS/heat-pump routes.

Superheated-steam drying + MVR/heat pump (emerging electric rival)

How it works: Closed steam loop dries material; mechanical vapor recompression recovers latent heat electrically (Fraunhofer LowCarbDry).

How well it performs: Targets 2-4x lower energy than hot-air drying; heat-pump drying ~0.14-0.31 kWh/kg water; ~4000 h demo operation reported in H2020 DryFiciency.

Limits: TRL ~5-6, pilot scale; compressor capex; steam chemistry/corrosion; unproven on abrasive bulk minerals.

Microwave/RF dielectric drying (ProSeed)

How it works: 915/2450 MHz fields volumetrically and selectively heat water inside the solid; energy deposited in the product, not in air.

How well it performs: Measured on iron pellets: ~120 vs ~170 kWh/t product (-30%) and 6-13x faster than convective; drying efficiency up to 56-78% vs ~30% convective; tailings >50% energy cut. Modules 6-300 kW (SAIREM, Pueschner, Ferrite/IMS).

Limits: Low-dielectric-loss matrices dry poorly (silica sand DE ~40%); multi-MW scale-up unproven commercially; electricity costs ~3-5x gas per kWh — economics hinge on efficiency gain + carbon price.

Why our solution can be better

Microwave drying deposits energy directly in the water inside the solid instead of heating tonnes of air and losing it in the exhaust — measured 30-50% lower specific energy on iron pellets and tailings at pilot scale. Being fully electric, it exits EU ETS and combustion-permitting scope, can run on renewable PPAs for scope-1-zero product claims (Mondo Minerals and Nordic Mining already proved this buyer logic with conventional electric heating), and offers instant controllability for demand response. Incumbents are ~30-60% thermally efficient against a 0.63 kWh/kg latent floor, leaving real headroom; the open question is scale-up economics versus cheap gas.

Key assumptions

• Drying = ~10-25% of industrial energy consumption, and mineral dryers today are overwhelmingly gas/oil-fired (measured, literature/industry).

• Incumbent rotary/fluid-bed dryers use ~1.0-2.5 kWh_th per kg water vs 0.63 kWh/kg latent minimum, i.e. >=30% efficiency headroom for any better technology (measured).

• Microwave drying saves ~30-50% energy vs convective on iron pellets/tailings (measured, pilot); kaolin/CaCO3 dielectric response likely between silica sand (DE ~40%, poor) and iron ore/basalt (DE ~65-78%, good) — must be measured per product (assumed).

• A 50 t/h line drying 10%->0.5% moisture evaporates ~5 t/h water ~3-4 MW_th; commercial MW modules today top out ~300 kW, so ~10x array scale-up is needed — an engineering, not physics, gap (measured + assumed).

• Economics work where carbon-priced gas plus ETS (~EUR60-90/tCO2) meets renewable electricity <~EUR80-100/MWh; strongest in EU/Nordics, weakest on cheap coal gas in Asia (assumed).

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
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Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis